Indian Institute of Science Education and Research, Kolkata
Math Methods (PH2204)-Spring 2025
Solution of Class Test -I
Tutor: *P. K. Mohanty*, Dept. Phys. Sc. (DPS)

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Notation $x \equiv x(t)$, $\dot{x} = \frac{d}{dt}x(t) \equiv Dx(t)$, $\ddot{x} = \frac{d^2}{dt^2}x(t) \equiv D^2x(t)$, $i = \sqrt{-1}$

1. **(a)** Find the generic solution of the ordinary differential equation (ODE): $\ddot{x} - 6\dot{x} + 25x = 0$. **[5]**
   (b) Show that the above ODE does not have a solution for boundary conditions $x(0) = 1, x(\pi) = 2$. **[5]**
   (c) Find the solution for the boundary conditions $x(0) = 1, \dot{x}(0) = -4$. **[5]**

Answer :

(a) We have the equation

$$\ddot{x} - 6\dot{x} + 25x = 0$$

It's characteristic equation is given by

$$\lambda^2 - 6\lambda + 25 = 0$$
$$\implies \lambda = \frac{-(-6) \pm \sqrt{(-6)^2 - 4 \times 25 \times 1}}{2}$$
$$\implies \lambda = 3 \pm 4i$$

Hence the generic solution is given by

$$\begin{aligned}
x(t) &= c_1 e^{(3+4i)t} + c_2 e^{(3-4i)t} \\
\implies x(t) &= e^{3t}(c_1 e^{4it} + c_2 e^{-4it}) \\
\implies x(t) &= e^{3t}(c_1(\cos(4t) + i\sin(4t)) + c_2(\cos(4t) - i\sin(4t))) \\
\implies x(t) &= e^{3t}((c_1 + c_2)\cos(4t) + i(c_1 - c_2)\sin(4t)) \\
\implies x(t) &= e^{3t}(A\cos(4t) + B\sin(4t))
\end{aligned}$$

Where A and B are constants.

(b) Given that $x(0) = 1$ and $x(\pi) = 2$. We have the general solution

$$x(t) = e^{3t}(A\cos(4t) + B\sin(4t))$$

Using $x(0) = 1$ we get

$$x(0) = e^{3\times 0}(A\cos(4 \times 0) + B\sin(4 \times 0)) = 1aa$$
$$\implies A = 1$$

Now using $x(\pi) = 2$ we get

$$x(\pi) = e^{3\pi}(A\cos(4\pi) + B\sin(4\pi)) = 2$$
$$\implies A = 2e^{-3\pi}$$

which is inconsistent, hence we cannot have a solution.

(c) Given that $x(0) = 1$ and $\dot{x}(0) = -4$. We have the general solution

$$x(t) = e^{3t}(A\cos(4t) + B\sin(4t))$$

Using $x(0) = 1$ we get

$$x(0) = e^{3\times 0}(A\cos(4 \times 0) + B\sin(4 \times 0)) = 1$$
$$\implies A = 1$$

Now using $\dot{x}(0) = -4$

$$\begin{aligned}
x(t) &= e^{3t}(A\cos(4t) + B\sin(4t)) \\
\implies \dot{x}(t) &= 3e^{3t}(A\cos(4t) + B\sin(4t)) + e^{3t}(-4A\sin(4t) + 4B\cos(4t)) \\
\implies \dot{x}(t) &= e^{3t}((3A+4B)\cos(4t) + (3B-4A)\sin(4t)) \\
\implies \dot{x}(0) &= e^{3\times 0}((3A+4B)\cos(4\times 0) + (3B-4A)\sin(4\times 0)) = -4
\end{aligned}$$

$$\implies 3A + 4B = -4$$

Since $A = 1$, we get

$$B = \frac{-7}{4}$$

Hence the solution for the given boundary conditions is

$$x(t) = e^{3t}\left(\cos(4t) - \frac{7}{4}\sin(4t)\right)$$

2. Consider the matrix $M = \begin{pmatrix} x & a & a & 1 \\ a & x & b & 1 \\ a & b & x & 1 \\ a & b & c & 1 \end{pmatrix}$. Find the values of x that solves $\det(M) = 0$. **[10]**

Answer :
By inspection one can see

- $x = a$ makes first column a times fourth column.
- $x = b$ makes second and third row identical
- $x = c$ makes third and fourth row identical

In all these cases, $\det M = 0$.

3. **(a)** Show that $\frac{1}{D^2+a^2}\sin(\omega t) = \frac{1}{-\omega^2+a^2}\sin(\omega t)$. **[10]**
(b) What happens when $a = \omega$? **[5]**

Answer :
(a) Note that $De^{iat} = iae^{iat}$, therefore (if $f(ia) \neq 0$, where f is a polynomial), $f(D)e^{iat} = f(ia)e^{iat} \implies f(D)(\frac{e^{iat}}{f(ia)}) = e^{iat}$. (In our case $f(x) = x^2+a^2$), Thus, for solving equations of the form $f(D)P(t) = e^{iat}$, if $f(ia) \neq 0$, we have.

$$P(t) = \frac{e^{iat}}{f(ia)}$$
$$\implies \frac{e^{iat}}{f(D)} = P(t) = \frac{e^{iat}}{f(ia)}$$

Using this we shall solve for $f(D)P(t) = (D^2+a^2)P(t) = sin(\omega t)$. We shall in turn solve for $f(D)P_1(t) = e^{i\omega t}$ and $f(D)P_2(t) = e^{-i\omega t}$.

$$\frac{1}{2i}(f(D)P_1(t) - f(D)P_2(t) = \frac{1}{2i}(e^{i\omega t} - e^{-i\omega t}) = \sin(\omega t)$$
$$\implies f(D)\big(\frac{1}{2i}(P_1(t) - P_2(t)\big) = \sin(\omega t)$$

which shows $\frac{1}{2i}(P_1(x) - P_2(x))$ is our required solution. By previous arguments, since $f(i\omega) = f(-i\omega) = -\omega^2 + a^2 \neq 0$ unless $a = \omega$, therefore,

$$P_1(t) = \frac{e^{i\omega t}}{-\omega^2 + a^2}$$
$$P_2(t) = \frac{e^{-i\omega t}}{-\omega^2 + a^2}$$
$$\frac{1}{D^2 + a^2}\sin(\omega t) = \frac{1}{2i}(P_1(t) - P_2(t)) = \frac{1}{-\omega^2 + a^2}\left(\frac{e^{i\omega t} - e^{-i\omega t}}{2i}\right) = \frac{\sin(\omega t)}{-\omega^2 + a^2}$$

(b) Now if $f(ia) = 0$ but $f'(ia) \neq 0$ then it follows,

$$f(x) = Q(x)(x - ia) \implies f'(x) = Q'(x)(x - ia) + Q(x)$$
$$f'(ia) = Q(ia)$$

Note

$$f(D)(e^{iat}t) = Q(D)(D - ia)(e^{iat}t)$$
$$= Q(D)((ia - ia)e^{iat}t + e^{iat}) = Q(D)e^{iat} = Q(ia)e^{iat} = f'(ia)e^{iat}$$

Therefore for solving $f(D)P(t) = e^{iat}$ where $f(ia) = 0$ but $f'(ia) \neq 0$,

$$\frac{e^{iat}}{f(D)} = P(t) = \frac{te^{iat}}{f'(ia)}$$

In the case when $a = \omega$, $f(i\omega) = f(-i\omega) = 0$, but $f'(i\omega) = 2i\omega$ and $f'(-i\omega) = -2i\omega$. In an analogous argument to the previous part, we shall solve for $f(D)P_1(t) = e^{i\omega t}$ and $f(D)P_2(t) = e^{-i\omega t}$. But in this case

$$P_1(t) = \frac{te^{i\omega t}}{2i\omega}$$
$$P_2(t) = \frac{te^{-i\omega t}}{-2i\omega}$$
$$\frac{1}{D^2 + a^2}\sin(\omega t) = \frac{1}{2i}(P_1(t) - P_2(t)) = \frac{t}{2i\omega}\left(\frac{e^{i\omega t} + e^{-i\omega t}}{2i}\right) = \frac{t\cos(\omega t)}{2i^2\omega} = -t\frac{\cos(\omega t)}{2\omega}$$

4. **(a)** The general solution of x in ODE $\ddot{x} + \beta\dot{x} + \gamma x = 0$ can either (A) diverge ($x(t) \to \infty$ as $t \to \infty$), (B) Settles ($x(t) \to 0$ as $t \to \infty$) or (C) oscillates. Find the regions in β -γ plane for all three cases. **[15]**
(b) Does this result depend on the choice of initial conditions? Explain, why, or why not. **[5]**

Answer :
(a) $\ddot{x} + \beta\dot{x} + \gamma x = 0 \implies (D - \lambda_+)(D - \lambda_-)x = 0$, where $\lambda_\pm$ are solution of the quadratic equation $D^2 + \beta D + \gamma = 0$. Thus,

$$\lambda_\pm = \frac{1}{2}\left(-\beta \pm \sqrt{\beta^2 - 4\gamma}\right). \tag{1}$$

Thus the general solution is $x(t) = c_1 e^{\lambda_- t} + c_2 e^{\lambda_+ t}$.

$$x(t) = e^{-\beta t/2}\left(c_1 e^{t/2\sqrt{\beta^2 - 4\gamma}} + c_2 e^{-t/2\sqrt{\beta^2 - 4\gamma}}\right)$$

When both $c_1, c_2 \neq 0$, the fate of $x(t)$ for large t depends on the sign of $\Re(\lambda_\pm)$. It diverges (i.e. $x(t) \to \infty$) if one of $\Re(\lambda_-)$ or $\Re(\lambda_+)$ or both are +ve, $x(t) \to 0$ if both $\Re(\lambda_+), \Re(\lambda_+)$ are −ve, and it oscillates when $\Re(\lambda_+) = 0 = \Re(\lambda_-)$.

- For $\gamma < 0$: $\sqrt{\beta^2 - 4\gamma}$ is real, +ve and larger than β. Thus λ_+ is real and positive $\implies$ (A).

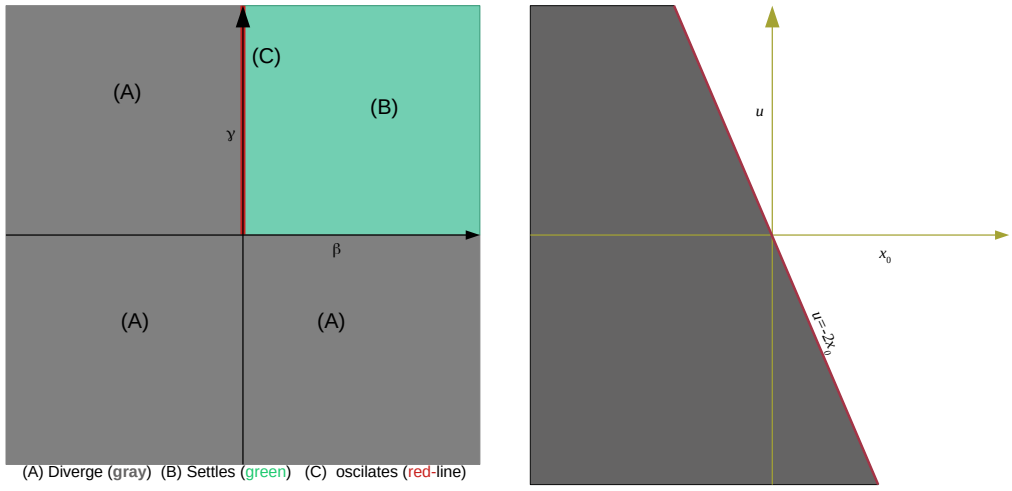

Figure 1: (left) Figure for Q4. (right) Figure for Q6.

- For $\gamma > 0$: Note in this case either $\sqrt{\beta^2 - 4\gamma} < \beta$ if $\beta^2 > 4\gamma$ or it is imaginary. If $\beta^2 = 4\gamma$, general solution is $e^{-\beta t/2}(c_1 + c_2 t)$. But still the exponential term is dominating. Therefore just the sign of β plays the important role in determining the nature of the solution as t $\to \infty$.
  - $\beta > 0 : e^{-\beta t} \to 0$ as $t \to \infty \implies$ (B)
  - $\beta = 0 : \lambda_\pm = \pm\frac{i}{2}\sqrt{|\gamma|} \implies$ (C)
  - $\beta < 0 : e^{-\beta t/2} = e^{|\beta|t/2} \to \infty$ as $t \to \infty \implies$ (A)
- $\gamma = 0$:
  - $\beta > 0$: General solution will be $c_1 e^{-\beta t} + c_2$, but $e^{-\beta t} \to 0$ as $t \to \infty$ so the system will settle at c_2 as $t \to \infty$ which goes to 0 if $c_2 = 0 \implies$ (B)
  - $\beta = 0$: General solution will be $c_1 t + c_2 \implies$ (A)
  - $\beta < 0$: General solution will be $c_1 e^{-\beta t} + c_2$, but $e^{-\beta t} = e^{|\beta|t} \to \infty$ as $t \to \infty \implies$ (A)

Figure 1 (left) divides β-γ plane into regions of (A),(B),(C).
(b) Yes, the results depend on initial condition. The above analysis in (a) assumes that the initial condition is such that both $c_{1,2} \neq 0$. But for initial conditions that results in one of $c_{1,2}$ is zero can lead to different results depending on initial condition. In cases where both $\lambda_\pm$ are real but one of them is positive and the other one is negative (which means their product $\lambda_+\lambda_=\gamma$ is $-$ve), then the final result depends on $c_{1,2}$. For example, when $\beta > 0$, $\gamma < 0 \implies \lambda_+$ is $+$ve and λ_- is $-$ve: here, if $c_1 = 0$ corresponds to (A), whereas $c_2 = 0$ corresponds to (B).

5. **(a)** Let L be a positive real number. Show that the ODE $\ddot{x} + \lambda x = 0$ with boundary condition $x(0) = 0 = x(L)$ has only the trivial solution $x = 0$, when $\lambda \leq 0$.**[5]**
(b) For $\lambda > 0$ find the solution explicitly. **[10]**

Answer :

- **(a)**
  $\lambda = 0$

$$\begin{aligned}
\ddot{x} + \lambda x &= 0 \\
\ddot{x} &= 0 \\
x(t) &= at + b \\
x(0) = 0 &\implies b = 0 \\
x(L) = 0 &\implies a \cdot L = 0 \implies a = 0 \\
x(t) &= 0
\end{aligned}$$

For $\lambda < 0$

$$\begin{aligned}
\ddot{x} + \lambda x &= 0 \\
\ddot{x} &= -\lambda x \\
-\lambda = k^2 > 0 &\implies \ddot{x} = k^2 x \\
x(t) &= c_1 e^{kt} + c_2 e^{-kt} \\
x(0) = c_1 + c_2 = 0 &\implies c_1 = -c_2 = c \\
x(t) &= c\left(e^{kt} - e^{-kt}\right) \\
x(L) = 0 &\implies c\left(e^{kL} - e^{-kL}\right) = 0 \implies ce^{-kL}\left(e^{2kL} - 1\right) = 0
\end{aligned}$$

Since $L > 0$, $e^{2kL} \neq 1$. Therefore, $x(t) = 0$.

- **(b)** For $\lambda > 0$

$$\begin{aligned}
\ddot{x} &= -\lambda x \\
\lambda &= k^2 \\
\ddot{x} &= -k^2 x \\
x(t) &= A\cos(kt) + B\sin(kt) \\
x(0) = 0 &\implies A = 0 \\
x(L) = 0 &\implies B\sin(kL) = 0 \\
kL = n\pi &\implies k = \frac{n\pi}{L}
\end{aligned}$$

The general solution is:$x(t) = B\sin\left(\frac{n\pi}{L}t\right)$

6. **(a)** Newton's equation of motion of a particle of mass $m = 1$ unit is given by $\ddot{x} = -2x - 3\dot{x}$. If starts from $x = x_0$ with velocity $\dot{x} = u$, show that in $t \to \infty$ it always comes to rest at $x = 0$. **[5]**
(b) For what values of x_0, u it approaches the resting point $x = 0$ from left? Shade the region in x_0, u plane. **[10]**
(c) If at $t = 0$, it starts moving right with velocity $u = 1$ from $x_0 = 1$, when is the first time it starts moving to left. **[10]**

Answer :
(a) $\ddot{x} = -2x - 3\dot{x} \implies (D^2 + 3D + 2)x = 0$
The roots are $\lambda = -2, -1$
$x(t) = C_1 e^{-2t} + C_2 e^{-t}$
For $x(0) = x_0$ and $\dot{x}(0) = u$ so $C_1 = -(x_0 + u)$ & $C_2 = (2x_0 + u)$
As $t \to \infty$;, this implies that $x \to 0$;

(b) For large t C_2 is effective (as e^{-2t} decays faster than e^{-2t}) and the particle is coming from the left it means $x(t) < 0; C_2 < 0; 2x_0 + u < 0$ the region in x_0 -u plane is shown in Figure 1(right).
(c) By putting $u = 1$ and $x_0 = 1$
$\dot{x}(t) = 4e^{-2t} + 3e^{-t}$
As it is moving from left $v(t) < 0$
It will only happen when $t > \ln(4/3)$

7. **(a)** Find a 3×3 matrix that converts the vector $\langle 1| = (1, 0, 0)$ to $\langle 3| = (0, 0, 1)$. **[5]**
(b) Show that $e^{i\sigma} = \cos(\sigma) + i\sin(\sigma)$, where σ is a 2×2 matrix $\sigma = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Hint: use $e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \ldots$ $\sin(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \ldots$. $\cos(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 + \ldots$. Note that $\sigma^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, the 2×2 identity matrix **[10]**.

(c) Write $\cos(t\sigma)$ as a 2×2 matrix $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$. Find the elements explicity. **[10]**

(d) Using matrix-method, solve $\ddot{x} = x$ with boundary condition $x(0) = 0, x(1) = 1$. **[10]**

(e) Find a, b, c so that the symm. matrix $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ gives same eigenvalues as in $B = \begin{pmatrix} 3 & 8 \\ 2 & 0 \end{pmatrix}$. **[5]**

Answer :

(a)

$M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$; by inspection $(1,0,0)\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = (0,0,1)$.

(b) $\sin(A) = A - \frac{1}{3!}A^3 + \frac{1}{5!}A^5 - \frac{1}{7!}A^7 + \ldots$; and $\cos(A) = I - \frac{1}{2!}A^2 + \frac{1}{4!}A^4 - \frac{1}{6!}A^6$
Now,

$$\begin{aligned} e^{iA} &= I + \frac{1}{2!}(iA)^2 + \frac{1}{3!}(iA)^3 + \frac{1}{4!}(iA)^4 + \frac{1}{5!}(iA)^5 \ldots \\ &= (I - \frac{1}{2!}A^2 + \frac{1}{4!}A^4 - \ldots) + i(A - \frac{1}{3!}A^3 + \frac{1}{5!}A^5 - \frac{1}{7!}A^7 + \ldots \\ &= \cos(A) + i\sin(A), \end{aligned}$$

if A^k is not a null-matrix for any interger $k > 0$.

Since $\sigma^2 = I$ we have $\sigma^{2n} = I$ and $\sigma^{2n+1} = \sigma$ for any integer n. Thus, $e^{i\sigma} = \cos(\sigma) + i\sin(\sigma)$.

(c) Since $\cos(t\sigma)$ contains all even-power of σ, and $\sigma^{2n} = I$ for any integer n,

$$\cos(t\sigma) = (1 - \frac{1}{2!}t^2 + \frac{1}{4!}t^4 + \ldots)I = \cos(t)I = \begin{pmatrix} \cos(t) & 0 \\ 0 & \cos(t) \end{pmatrix}. \tag{2}$$

(d) Say $\dot{x} = v$ then the equation we have is $\dot{v} = x$; and thus

$$\begin{aligned} &\frac{d}{dt}\begin{pmatrix} x \\ v \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ v \end{pmatrix} = \sigma\begin{pmatrix} x \\ v \end{pmatrix} \\ \Longrightarrow \quad &\begin{pmatrix} x \\ v \end{pmatrix}_t = e^{t\sigma}\begin{pmatrix} x \\ v \end{pmatrix}_0 = e^{t\sigma}(c_+|e_+\rangle + c_-|e_-\rangle) = c_+e^{\lambda_+ t}|e_+\rangle + c_-e^{\lambda_- t}|e_-\rangle \end{aligned} \tag{3}$$

where $|e_\pm\rangle = \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$ are eigen vectors of σ with eigenvalues $\lambda_\pm = \pm 1$. Thus

$$x(t) = c_+e^{\lambda_+ t} + c_-e^{\lambda_- t} \quad \text{and} \quad v(t) = c_+e^{\lambda_+ t} - c_-e^{\lambda_- t}. \tag{4}$$

The boundary condition $x(0) = 0$ gives $0 = c_+ + c_-$ and $x(1) = 0$ gives $1 = c_+e + c_-e^{-1}$, which can be solved to $c_+ = \frac{1}{e - e^{-1}} = -c_-$. Finally,

$$x(t) = \frac{1}{e - e^{-1}}\left(e^{\lambda_+ t} - e^{\lambda_- t}\right) \quad \text{and} \quad v(t) = \frac{1}{e - e^{-1}}\left(e^{\lambda_+ t} + e^{\lambda_- t}\right). \tag{5}$$

(e) Since $trace(B) = \lambda_1 + \lambda_2 = 3$ and $\det(B) = \lambda_1\lambda_2 = -16$, matrix A must have $trace(A) = 3$ and $\det(A) = -16$. A simple choice is $A = \begin{pmatrix} 3 & 4 \\ 4 & 0 \end{pmatrix}$.

Thus any 2×2 matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ is isospectral to a symmetric matrix $\begin{pmatrix} p & \sqrt{qr} \\ \sqrt{qr} & s \end{pmatrix}$